Rayyan Omar

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Technical Skills

Languages: Java, Python, JavaScript, SQL, TypeScript, HTML/CSS
Frameworks: FastAPI, React, Node.js, Spring Boot, JUnit, Express, Snowflake API
Developer Tools: MongoDB, Git, GitHub Actions, CI/CD, Docker, Event-Driven Architecture, Multi-Agent Systems, OpenRouter

Experience

Materials Research Student (Co-op) Sep 2023 – Jun 2024
National Research Council (NRC) – Aerospace Center *Ottawa, ON*

- Validated high-temperature material resistance for over 100+ aerospace samples by designing and executing rigorous testing protocols, ensuring compliance with strict safety parameters.
- Enhanced material degradation analysis accuracy by 30% through the generation of high-resolution 3D depth maps and microscopic imaging workflows.
- Standardized testing operations across the facility by authoring a 25-page technical safety manual adopted by all subsequent researchers.

Projects

AeroGuard (A.E.G.I.S. Garden) | *FastAPI, Snowflake, MongoDB, Google Gemini* 2026

- Built an AI-powered security operations (SOC) platform, winning **First Place for Best Use of Snowflake** at HackConcordia by effectively eliminating alert fatigue.
- Reduced incident triage decision time to under 60 seconds by integrating Google Gemini LLM APIs for prompt-based, explainable alert generation.
- Scaled backend data ingestion by architecting parallel flows using MongoDB for state management and Snowflake for deep telemetry analytics.

Chronos Cloud: AI Smart City Command Center | *FastAPI, Python, 14 APIs, 7 AI Agents* 2026

- Engineered a real-time command center by orchestrating an event-driven architecture with 14 APIs and 7 specialized AI agents for multi-step tasks.
- Reduced complex decision-making latency for city operators by over 80% (from hours to minutes) by deploying agents to correlate cascading risks across transit and traffic systems.
- Ensured AI safety and 100% auditability by implementing a human-in-the-loop verification UI that evaluates and logs every generated AI assumption and telemetry-verified outcome.

ParkingSporter | *Java, Spring Boot, React, Android Studio* 2025

- Architected a full-stack mobile application backend within a 36-hour sprint by integrating a Java/Spring Boot environment with a React frontend.
- Achieved <500ms response latency for data processing by building a scalable REST API that interfaces with external APIs for high-speed incident detection.

Leadership

Co-VP of Logistics 2026
Hack the Hill Hackathon *Ottawa, ON*

- Directed end-to-end event operations by orchestrating check-ins, ticketing pipelines, and real-time problem-solving protocols for 900+ participants, significantly reducing entry queue times.
- Facilitated technical education at scale by organizing and managing Full-Stack Development and AI/ML workshop panels, delivering specialized curriculum to an audience of 500+ attendees.
- Sustained continuous 36-hour event operations by negotiating and managing large-scale catering services for 900+ attendees and securing centralized overnight accommodations for the organizing team.

Education

University of Ottawa Ottawa, ON
BASc in Software Engineering (CO-OP) *March 2028*

Colonel By Secondary School (IB Program) Ottawa, ON
High School Diploma *Jun 2024*